

Matrix inequalities and norms

The complex Lu–Wenzel spectral conjecture

Difficulty: extreme **Importance:** interesting to the community**Status:** Partially resolved**Provenance:** explicit conjecture**Rating rationale:** Uniform control of the full commutator spectrum is a major strengthening of established norm bounds; it connects Sylvester operators with geometric matrix inequalities.

Problem statement

For every integer $n \geq 2$ and $X \in \mathbb{C}^{n \times n}$ with $\|X\|_F = 1$, consider the complex-linear operator

$$T_X : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{n \times n}, \quad T_X(Y) = [X^*, [X, Y]], \quad [P, Q] = PQ - QP.$$

Use the inner product $\langle Y, Z \rangle_F = \operatorname{tr}(Y^*Z)$. The operator is Hermitian positive semidefinite; list its n^2 eigenvalues, including multiplicities, as $\lambda_1(T_X) \geq \dots \geq \lambda_{n^2}(T_X) \geq 0$. Is

$$\sum_{i=1}^{2k} \lambda_i(T_X) \leq 2k + 2 \quad \text{for every integer } 1 \leq k \leq \lfloor n^2/2 \rfloor?$$

Relevance

This bounds sums of squared singular values of the commutator map $Y \mapsto XY - YX$. Such a map is a basic Sylvester operator. The conjecture strengthens a bound on its largest singular value to simultaneous bounds on its singular spectrum.

References

1. J. Ge, F. Li, Z. Lu, and Y. Zhou, *On some conjectures by Lu and Wenzel*, arXiv:1908.06624v1, Conjecture 7; Theorems 1.1–1.2 and §4. [Primary manuscript](#).
2. J. Ge, F. Li, Z. Tang, and Y. Zhou, *A survey on the DDVV-type inequalities*, Advances in Mathematics (China) 53 (2024), 449–467: published Conjecture 4.6 and Theorems 4.1–4.2, pp.463–464. These are Conjecture 4.14 and Theorems 4.17–4.18 in [arXiv:2402.01085v1](#). [Published PDF](#).
3. Z. Liu, *The Lu–Wenzel Spectral Conjecture for Square-Zero Matrices*, Zenodo record 21693383, abstract. [Author’s preprint record](#).

Status check — 2026-09-10

Both arXiv sources remain v1. The published survey retains the conjecture and records the normal, rank-one, and $n = 2, 3$ cases. Liu’s record claims the additional class $X^2 = 0$ and scalar translates; its abstract does not claim the unrestricted assertion. That record’s displayed publication date and July 2026 creation metadata differ, so no priority date is inferred from it. Searches used *Lu–Wenzel conjecture proof 2026*, *Lu Wenzel conjecture solved*, and *Lu–Wenzel spectral conjecture*. No general resolution was located. Equivalent majorization and fundamental-commutator formulations, and the real restriction, are grouped in this entry.

Audit update (2026-09-10): Rechecked the 2024 survey’s Conjecture 4.14 and special-case discussion, and searched for later Lu–Wenzel spectral results. The normal, rank-one and low-order cases are within the displayed target; the square-zero preprint claims only another restricted class. This is a bounded literature check, not a proof that no solution exists.